

**Time:** 5 minutes**Outcome:** Calculate CPI and inflation using a market basket.**Difficulty:**

Intermediate

## CPI and Inflation Measurement



### THE CORE IDEA

The consumer price index compares the current cost of a fixed consumer basket with its base-year cost. The index equals the current basket cost divided by the base-year basket cost, multiplied by one hundred. Inflation is the percentage change in the index between periods. Holding basket quantities fixed isolates price change from changes in what consumers buy.



### HOW TO RECOGNIZE IT

- Identify the fixed basket, its base-year cost, and its current-year cost.
- Compute the price index by comparing the current basket cost with the base-year basket cost.
- Measure inflation as the percentage change in the index from one period to the next.
- Keep basket quantities fixed when calculating the CPI for different years.



### WATCH OUT

Do not use the wrong basket cost or forget to multiply by 100. Divide the current cost of the fixed basket by its base-year cost; the index can rise or fall as prices change.



### WORKED EXAMPLE: CALCULATING CPI AND INFLATION

The CPI rises from 125 to 150. The inflation rate is  $(150 - 125)/125 \times 100 = 20\%$ . In a separate basket, base-year cost is  $4(\$10) + 2(\$20) = \$80$  and current cost is  $4(\$12) + 2(\$25) = \$98$ , so CPI is  $(\$98/\$80) \times 100 = 122.5$ .



### CHECK YOURSELF

A basket costs \$400 in the base year and \$500 today. What is today's CPI?

**READY?**

Return to the game and master this concept.